

Minimal file-based note system for messy scientists

Static site generator for hyperlinked notes

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Joseph Wilson & Mete Polat

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**1 There are many ways to
take notes in this modern
world**

Existing solutions



- Obsidian, <http://obsidian.md/>
 - Markdown-based
 - Extensible
- Notion, <http://notion.so>
 - All-in-one organisational tool
- “Zettelkasten” gardens, <https://notes.andymatuschak.org>
 - Highly hyperlinked webs of small markdown files
 - Beautiful web frontend
- Evernote, <https://evernote.com/>

What I want



- Format agnostic
 - Markdown files for quick notes
 - Jupyter notebooks with code and data
 - Word documents
 - LaTeX or Typst PDFs
 - Screenshots or image files with explanations
- Easy hyperlinking and cross-referencing
 - To other notes
 - Cite published material
- Free hosting, easy sharing
- Don't tell me how I should take and organise my notes!

2 The simplest useful solution

Just store notes as files



A screenshot of a macOS Finder window titled "notes". The window displays a file hierarchy with folders and files. The files are listed in a table with columns for Name, Date Modified, and Size.

Name	Date Modified	Size
▼ Hack Day talk	Today at 16:32	--
about-this-project.note.pdf	Today at 16:32	357 KB
about-this-project.note.typ	Today at 16:32	1 KB
files.png	Today at 16:30	458 KB
joe-primes.note.pdf	22 Oct 2024 at 21:58	155 KB
joe-primes.note.typ	23 Oct 2024 at 01:34	433 bytes
> Linear algebra	Yesterday at 16:30	--
▼ Statistics	23 Oct 2024 at 14:19	--
central-limit-theorem.note.pdf	16 Oct 2024 at 18:00	161 KB
central-limit-theorem.note.typ	16 Oct 2024 at 17:58	2 KB
> Gaussian distributions	22 Oct 2024 at 17:02	--
poisson-distribution.note.pdf	22 Aug 2024 at 23:22	21 KB
poisson-distribution.note.tex	22 Aug 2024 at 23:22	3 KB
▼ Probabilistic Machine Learning 4f13 Michaelmas 2024	Yesterday at 16:31	--
gaussian-processes.note.html	Yesterday at 16:31	901 KB
gaussian-processes.note.jl	Yesterday at 00:06	50 KB
variance-of-sum.note.pdf	16 Oct 2024 at 17:11	151 KB
variance-of-sum.note.typ	16 Oct 2024 at 17:11	858 bytes

Just store notes as files (ii)



Mark files as notes with `.note.` in filename.

Notes can have multiple files:

- `A.note.tex` + `A.note.pdf` = A LaTeX PDF
- `B.note.typ` + `B.note.pdf` = A Typst PDF
- `C.note.md` = A standalone markdown document
- `D.note.jl` = A raw Julia script
- `E.note.ipynb` + `E.note.html` = A Jupyter notebook with exported HTML
- `F.note.jl` + `F.note.html` = A Pluto.jl notebook
- `G.note.png` + `G.note.txt` = A picture with words

(All this is currently hard-coded into the build script)

Make simple static site to host them

Put your notes in a Git repo.

Add a GitHub workflow to run this action:

<https://github.com/Jollywatt/notes-deploy-action>

When pushed, a static site is built and hosted on GitHub pages:

<https://jollywatt.github.io/notes/>

Advantages



- Notes of all different sorts can live together
- Notes get permalinks so you can link to them and cross-reference
- To add a different format, the build script can be extended
- You can build and serve the site locally as an alternative way to navigate your files
- You get free hosting if you want your notes to be public

Disadvantages



- Currently super basic and breakable
- Doesn't work with notes that you couldn't easily put on a Git repo (Google Docs, Apple Notes)

3 Next steps

Integration with Typst



To cite other notes, you can write, e.g.:

```
This was proven in #link("https://jollywatt.github.io/notes/
proof-of-thing")[this note].
```

With a simple Typst package that hooks into metadata produced by the build script, you can write:

```
#import "@local/notes:0.1.0"
#show: notes.style
```

```
This was proven in @proof-of-thing[this note].
```

Adding HTML renderers for different formats.....

- Renderer for plain markdown files
- A format for adding words to pictures, e.g., `idea.note.png` + `idea.note.txt` could produce an HTML page with both side-by-side.